Grading Rubric

Lab 1: Introduction to MATLAB & Simulink (Total Marks: 20%)

Assessment Criteria	Excellent (100%)	Good (75%)	Satisfactory (50%)	Poor (25%)/Incomplete	Marks Allocated
MATLAB Code Accuracy	Code runs flawlessly; correct TF implementation.	Minor errors (e.g., misplaced labels).	Major errors (e.g., wrong TF formula).	Code fails to run or irrelevant.	/5%
Modified Transfer Function	Successfully altered parameters; clear rationale.	Modified but no explanation.	Partial modification (e.g., only numerator).	No modification attempted.	/3%
Step Response Plot	Professional plot (labels, title, grid).	Missing 1-2 elements (e.g., no grid).	Missing >2 elements or incorrect plot.	No plot submitted.	/4%
Analysis (Rise/Settling Time)	Correct `stepinfo()` usage; detailed comparison.	Minor calculation errors.	Incomplete analysis (e.g., only rise time).	No analysis provided.	/5%
Report Clarity	Neat, well-structured, answers explained.	Minor formatting issues.	Disorganized or lacks explanations.	Unsubmitted or illegible.	/3%

Lab 2: Transient Response of Second-Order Systems (Total Marks: 20%)

Assessment Criteria	Excellent (100%)	Good (75%)	Satisfactory (50%)	Poor (25%)/Incomplete	Marks Allocated
Plot (Multiple Responses)	All curves labeled; axes/title perfect.	Missing legend or minor labeling issues.	Incorrect responses or missing plots.	No plot submitted.	/5%
Modified ω_n Analysis	Clear explanation of speed vs. ω_n.	Correct but lacks depth.	Partial analysis (e.g., no comparison).	No analysis attempted.	/4%
Design Task (Correct ζ and ω_n)	Accurate design; meets specs.	Design works but minor deviations.	Incorrect parameters but partial effort.	No design attempted.	/6%
Overshoot Comparison	Manual and MATLAB results match; detailed.	Minor discrepancies in values.	Only one method provided.	No comparison.	/3%
Report Quality	Professional; screenshots embedded.	Minor formatting issues.	Poorly organized or incomplete.	Unsubmitted.	/2%

Lab 3: PID Controller Design (CLO2 Focus) (Total Marks: 30%)

Assessment Criteria	Excellent (100%)	Good (75%)	Satisfactory (50%)	Poor (25%)/Incomplete	Marks Allocated
PID Tuning Method	Correct use of `pidtune()` or manual tuning.	Used `pidtune()` but no manual adjustments.	Manual tuning only; no justification.	No tuning attempted.	/10%
Achieved Design Specs	Overshoot <10%; meets all requirements.	Overshoot 10-15% or minor deviations.	Overshoot >15% but stable.	Unstable or no improvement.	/10%
Before/After Plots	Clear comparison; labeled and annotated.	Missing annotations or titles.	Only one plot provided.	No plots submitted.	/5%
Tuning Explanation	Detailed rationale for chosen gains.	Brief or vague explanation.	Minimal explanation (e.g., 'it works').	No explanation.	/5%

Lab 4: Stability Analysis (Nyquist/Bode) - CLO3 (Total Marks: 30%)

Assessment Criteria	Excellent (100%)	Good (75%)	Satisfactory (50%)	Poor (25%)/Incomplete	Marks Allocated
Bode/Nyquist Plot	Correct plots with margins highlighted.	Minor errors (e.g., no margins).	Incorrect frequency range or scaling.	No plot submitted.	/10%
Stability Analysis	Accurate GM/PM; correct stability conclusion.	Correct GM/PM but unclear conclusion.	Errors in margin calculations.	No analysis.	/10%
Margins Calculation	Precise values; units included.	Minor rounding errors.	Incorrect values but partial effort.	No calculations.	/5%
Report Professionalism	Publication-ready; clear sections.	Minor formatting issues.	Disorganized or lacks screenshots.	Unsubmitted.	/5%